

MIDTERM ASSESSMENT FOR IT351 COMPUTER NETWORKS (SUMMER 2022-2023)

Assiut University



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Time Allowed: 1 hour 10 minutes

INTRODUCTION TO CANDIDATES:

- 1- This assessment paper contains **twenty (20)** questions.
- 2- The paper consists of two sections: **1) Multiple Choice Questions (MCQ)** and **2) Short Questions**. Each question is worth 1.5 mark and the maximum possible mark is **30**.
- 3- This is a **CLOSED BOOK** assessment.
- 4- Calculators are allowed.
- 4- You are **NOT** allowed to communicate with each other or search online for answers.

Points: 16.5/30

Student Information

1

Student Name (IN ARABIC) *

محمد زارع حسن زارع

2

Student ID *

162019130

Multiple Choice Questions (MCQ)

✓ **Correct** 1.5/1.5 Points

3

Which of the following layers doesn't appear in the Internet Protocol Stack?

☐ Application layer

☐ Network layer

☐ Transport layer

☒ Session layer

☐ Physical layer

✗ **Incorrect** 0/1.5 Points

4

Which of the following statements is true?

☐ Both the Internet and traditional telephone networks use packet-switching.

☐ Performance metrics such as end-to-end delay and throughput can be guaranteed in the Internet, if TCP is chosen as the transport layer protocol.

☐ In the Internet, packets from the same source always take the same path to reach destination.

☒ The Internet uses packet switching and traditional telephone networks uses circuit switching.

☒ None of the above

✗ Incorrect 0/1.5 Points

5

Which of the following statements is true about circuit switching?

- ☐ Resources are pre-allocated regardless of demand
- ☐ Packet transmission can use the full link bandwidth
- ☐ The path between a source and a destination is not fixed
- ☒ None of the above
- ☐ All of the above

✗ Incorrect 0/1.5 Points

6

Which of the following is present in both HTTP request line and status line?

- ☒ HTTP version number
- ☐ Request method
- ☐ Status code
- ☒ URL
- ☐ None of the above

✗ Incorrect 0/1.5 Points

7

Which of the following statements about DNS is TRUE?

- ☒ TLD servers store all the hostname to IP mappings of the Internet.
- ☒ Every Web server must have a canonical name.
- ☒ Local DNS server sometimes may provide out-of-date hostname-to-IP-address mapping.
- ☒ DNS server listens to TCP port 53.
- ☐ None of the above

✓ **Correct** 1.5/1.5 Points

8

A packet switch receives a packet and determines the outbound link to which the packet should be forwarded. When the packet arrives, x bits of the currently-being-transmitted packet have been transmitted on this outbound link and n other packets are waiting to be transmitted. Packets are transmitted continuously in order of arrival. Suppose all packets are L bits and the link rate is R bps.

What is the queuing delay?

- ☐ nL / R
- ☐ L / R
- ☐ $n(L - x) / R$
- ☐ $nL + (L - x) / R$
- ☒ $(nL + (L - x)) / R$
- ☐ $(nL + (x - L)) / R$

✓ **Correct** 1.5/1.5 Points

9

A TCP client sends a sequence of packets to a TCP server. Each packet is 1000 bytes and the link rate is 2 Mbps. The propagation delay between the client and server is 5 ms. Which of following is the best timeout value chosen by the TCP client?

- ☐ 7 ms
- ☐ 9 ms
- ☐ 11 ms
- ☐ 14 ms
- ☒ 16 ms

✓ **Correct** 1.5/1.5 Points

10

Consider the following Python code snippet.

Suppose no exception is raised when the above code snippet is executed, which of the following statements is TRUE?

```
...  
s1.bind('', 8080)  
s2, addr = s1.accept()  
...
```

- ☐ s1 is a UDP socket
- ☐ Server uses s1 to transmit application data to client
- ☒ s2 listens to port 8080
- ☐ s2 listens to a random port number assigned by operation system
- ☐ None of the above

✓ **Correct** 1.5/1.5 Points

11

Consider the transmission between a UDP sender and a UDP receiver. Which of the following will never happen? You may assume that the application riding on UDP doesn't implement any reliability mechanisms.

- ☐ UDP receiver fails to receive any packet from UDP sender.
- ☐ UDP receiver receives out-of-order packets from UDP sender.
- ☒ UDP receiver receives duplicate packets from UDP sender.
- ☐ Option 5 UDP receiver receives corrupted packets from UDP sender but fails to detect bit errors.
- ☐ None of the above

✓ **Correct** 1.5/1.5 Points

12

A browser requests to download index.html from a Web server. The HTTP response header received by the browser is shown below.

Which of the following statements is TRUE?

- ☐ In the HTTP request, browser has requested for a non-persistent connection.
- ☐ Suppose TCP header is 20 bytes, the length of the TCP segment containing the HTTP response is 1276 bytes.
- ☐ The index.html file received by the browser may have been corrupted during transmission.
- ☒ The Web server uses a cookie to keep the state information of the client.
- ☐ None of the above

✓ **Correct** 1.5/1.5 Points

13

In Go-back-N protocol, what are the size of the respective sender and receiver buffers required for a window size of N?

- ☐ 1; 1
- ☐ N - 1; 1
- ☐ N - 1; N - 1
- ☒ N; 1
- ☐ N; N

✓ **Correct** 1.5/1.5 Points

14

Consider a sender and a receiver communicating using Selective Repeat protocol. Every packet embeds a 4-bit sequence number field. Sender window size is 4. None of the packets shown in the following figure are corrupted packets. However, the third data packet and the second ACK packet are lost.

What is the sender window over the sequence number space at time t ?

- ☐ 12, 13, 14, 15
- ☐ 13, 14, 0, 1
- ☒ 13, 14, 15, 0
- ☐ 13, 14, 15, 16
- ☐ None of the above

✗ **Incorrect** 0/1.5 Points

15

A Selective Repeat sender just receives an ACK packet with ACK number 7. This ACK number falls within sender window which has the window size 3. Every data packet embeds a k -bit sequence number field (k is a constant unknown to you).

Which of the following definitely CANNOT be the sequence number of the next packet transmitted by the sender?

- ☐ 0
- ☐ 2
- ☒ 3
- ☐ 6
- ☒ 10

✓ **Correct** 1.5/1.5 Points

16

Consider the following pseudo-code for rdt 3.0 receiver.

Here, **seq** is the expected sequence number (either 0 or 1) by the receiver. The meaning of the rest function calls and variables are the same as those in FSM diagram. We say that the receiver waits forever, if it is blocked at the while loop, waiting to receive a packet that will never arrive. We say that the sender loops forever, if it repeatedly retransmits the same packet over and over again. Which of the following statements is TRUE?

- ☐ A single premature timeout is sufficient to cause the sender to loop forever.
- ☒ A single loss ACK packet is sufficient to cause the sender to loop forever.
- ☐ A single corrupted data packet is sufficient to cause the sender to loop forever.
- ☐ A single corrupted data packet is sufficient to cause the receiver to wait forever.
- ☐ None of the above

SHORT QUESTIONS

✓ **Correct** 1.5/1.5 Points

17

Suppose 5 data packets arrive at a TCP receiver. Each packet contains 10 bytes of application data. The sequence numbers of these 5 packets (shown in the order of arrival) are 50, 60, 90, 70 and 80 respectively. After receiving each packet, the TCP receiver sends an ACK packet. Suppose the first ACK packet has ACK number 60. What is the ACK number of the 3rd ACK packet?

70

✓ **Correct** 1.5/1.5 Points

18

What is the checksum (1's complement of the sum) of the following 3 bytes?

11110100
10010011
10011001

11011101

✗ **Incorrect** 0/1.5 Points

19

The following diagram shows two hosts X and Y communicating over a channel using TCP. X and Y are sending data to each other. Each TCP segment contains 100 bytes of data. None of the segments shown in the figure are retransmitted, out-of-order or corrupted packets. The first segment received by X has the sequence number 100 and ACK number 150.

What would be the sequence number (S) in the last segment received by Y?

200

Correct answers:

250

✗ Incorrect 0/1.5 Points

20

The following diagram shows two hosts X and Y communicating over a channel using TCP. X and Y are sending data to each other. Each TCP segment contains 100 bytes of data. None of the segments shown in the figure are retransmitted, out-of-order or corrupted packets. The first segment received by X has the sequence number 100 and ACK number 150.

What would be the ACK number (A) in the last segment received by Y?

250

Correct answers:

200

✗ Incorrect 0/1.5 Points

21

A source and a destination are separated by 3×10^5 kilometers and are connected by a direct link of 2 Kbps. The propagation speed over the link is 2×10^8 meters/second. The source sends 100 packets to the destination using RDT 2.2. Each packet is of 2×10^3 bits long.

Suppose ACK packets are of negligible size and transmission channel is perfectly reliable. What is the throughput (in bps) of transmission?

2000 bps

Correct answers:

500

✗ **Incorrect** 0/1.5 Points

22

A Web server stores a webpage that comprises a base HTML file and 2 images referenced by the base HTML file. The HTML file is 100 bytes and each image is 200 bytes. A client is connected to the Web server through a direct link of 1 Mbps. Propagation delay between the Web server and the client is 50 milliseconds.

The client downloads the webpage using persistent HTTP but without pipelining (i.e. the next HTTP request is sent after the response for the previous HTTP request is received). Assume HTTP header, TCP header and ACK packets are of negligible size, transmission channel is perfectly reliable, time to establish and close TCP connection can be ignored. How long (in milliseconds) does it take for the client to download the entire webpage from the Web server?

.154

Correct answers:

304

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